

DECODELABS INDUSTRIAL TRAINING KIT (BATCH 2026)

PROJECT 1: DATA CLEANING & PREPARATION — CHANGE LOG

INTERN DETAILS

Intern Name: Muhammad Raheem

Domain: Data Analytics

Project Milestone: Project 1 data clean (Foundation Phase)

Submission Date: July 5, 2026

TRANSFORMATION REPORT

1. Initial Data Audit:

Total Records: 1,200 rows with 14 columns.

Missing Values: Identified 309 missing (`NaN`) values strictly inside the `CouponCode` column.

Duplicates Check: Performed an integrity audit on `OrderID` and full rows; 0 duplicates found, ensuring strict identity uniqueness.

2. Imputation Strategy (Handling Gaps)

Action: Replaced all 309 missing values in `CouponCode` with the string standard `NO\_COUPON`.

Justification: Avoided listwise deletion to preserve statistical power and maintain the integrity of the 1,200 analytical records.

3. Standardization & Language Consistency

Date Formats: Verified and locked all entries in the `Date` column to the standard ****ISO 8601 format (YYYY-MM-DD)****.

Numeric Precision: Standardized both `UnitPrice` and `TotalPrice` columns to exact ****2 decimal places**** for professional business reporting.

4. Final Quality Gate Verification

Error Rate: Achieved a strict **'0% Error Rate'** on unique identifiers and formatting styles.

Status:The dataset has been transformed from raw noise into a production-ready **"Gold Standard"** reference for future dashboards and modeling.